

# Program 10

## Demonstrate Kubernetes Dashboard and CLI for Cluster Monitoring

### Project Structure:

Program-10/

→ deployment.yaml

→ service.yaml

### Step 1: Start Minikube with resources

minikube start --cpus=2 --memory=4096

```
1rv24mc012_akanksha@akanshasPC:~/Program9$ cd ..
1rv24mc012_akanksha@akanshasPC:~$ mkdir Program10
1rv24mc012_akanksha@akanshasPC:~$ cd Program10
1rv24mc012_akanksha@akanshasPC:~/Program10$ minikube start --cpus=2 --memory=4096
🐳 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
! You cannot change the memory size for an existing minikube cluster. Please first delete the cluster.
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📡 Pulling base image v0.0.48 ...
👉 Updating the running docker "minikube" container ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌞 Enabled addons: storage-provisioner, default-storageclass, metrics-server
🎉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

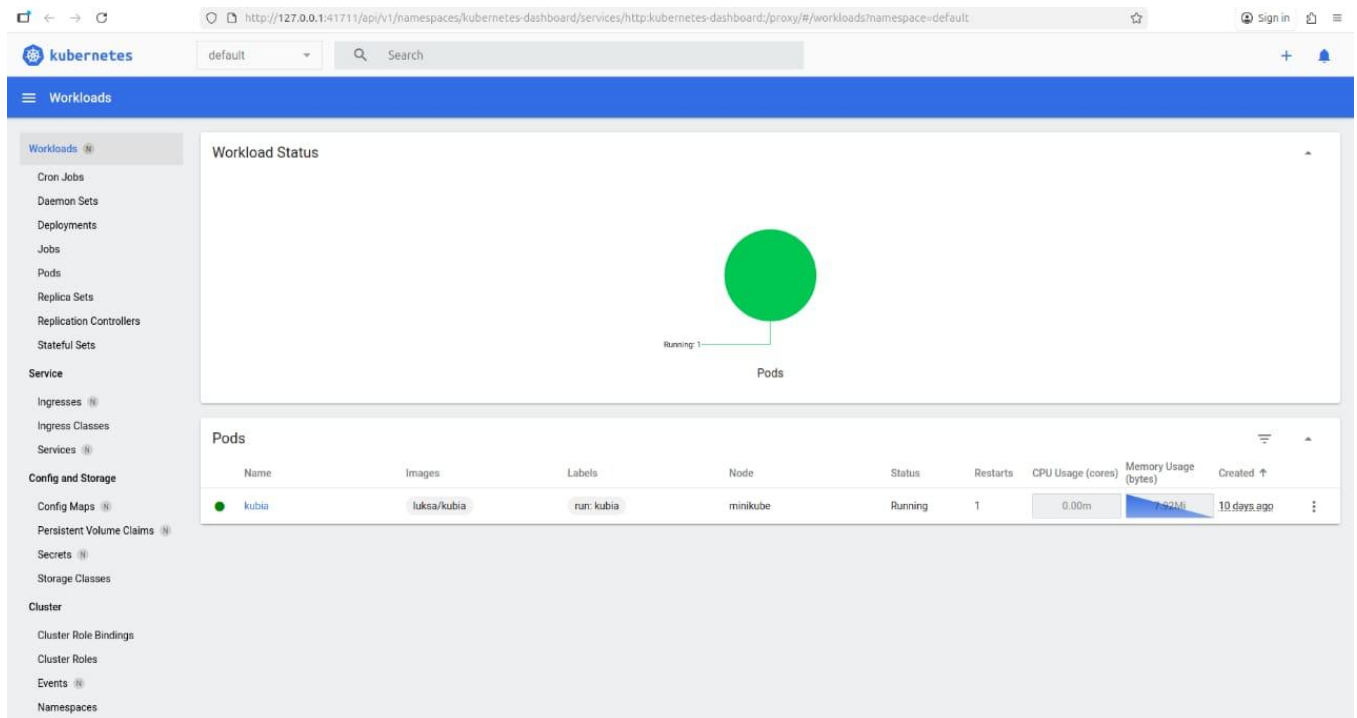
### Step 2: Open Kubernetes Dashboard

minikube dashboard

```
1rv24mc012_akanksha@akanshasPC:~/Program10$ minikube dashboard
🔧 Enabling dashboard ...
   ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
   ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

👉 Verifying dashboard health ...
🔧 Launching proxy ...
👉 Verifying proxy health ...
👉 Opening http://127.0.0.1:41711/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser
Gtk-Message: 09:47:41.161: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
^C
```



### Step 3: Create Deployment (deployment.yaml)

Create folder:

```
mkdir Program-10
```

```
cd Program-10
```

Create file:

```
nano deployment.yaml
```

Inside the file:

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: nginx-deployment
```

```
  labels:
```

```
    app: nginx
```

```
spec:
```

```
  replicas: 2
```

```
selector:
  matchLabels:
    app: nginx
template:
  metadata:
    labels:
      app: nginx
  spec:
    containers:
      - name: nginx
        image: nginx:latest
        ports:
          - containerPort: 80
```

### Apply:

```
kubectl apply -f deployment.yaml
```

### Verify:

```
kubectl get deployments
```

```
kubectl get pods
```

```
1rv24mc012_akanksha@akanshasPC:~/Program10$ nano deployment.yaml
1rv24mc012_akanksha@akanshasPC:~/Program10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
1rv24mc012_akanksha@akanshasPC:~/Program10$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    1/2     2            1           7s

1rv24mc012_akanksha@akanshasPC:~/Program10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
kubia                              1/1     Running   1 (4m45s ago)    10d
nginx-deployment-6f9664446b-r5fw7  1/1     Running   0           13s
nginx-deployment-6f9664446b-vs9r   1/1     Running   0           13s
```

## Step 4: Create Service (service.yaml)

### Create file:

nano service.yaml

### Inside the file:

apiVersion: v1

kind: Service

metadata:

name: nginx-service

spec:

selector:

app: nginx

ports:

- protocol: TCP

port: 80

targetPort: 80

type: NodePort

### Apply:

kubectl apply -f service.yaml

### Verify:

kubectl get svc

```
1rv24mc012_akanksha@akanshasPC:~/Program10$ nano service.yaml
1rv24mc012_akanksha@akanshasPC:~/Program10$ kubectl apply -f service.yaml
service/nginx-service created
1rv24mc012_akanksha@akanshasPC:~/Program10$ kubectl get svc
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)          AGE
kubernetes           ClusterIP   10.96.0.1     <none>       443/TCP          23d
kubia                NodePort    10.111.41.23  <none>       8080:30236/TCP   10d
nginx-service        NodePort    10.107.62.10  <none>       80:32284/TCP     5s
testdeploy           NodePort    10.102.248.183 <none>       80:32218/TCP     18d
```

## Step 5: Monitor using CLI

### Check pods

```
1rv24mc012_akanksha@akanshasPC:~/Program10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
kubia                               1/1     Running   1 (5m16s ago)   10d
nginx-deployment-6f9664446b-r5fw7  1/1     Running   0           44s
nginx-deployment-6f9664446b-vs9r   1/1     Running   0           44s
```

### Check nodes

### Check services

```
1rv24mc012_akanksha@akanshasPC:~/Program10$ kubectl get nodes
NAME        STATUS   ROLES    AGE   VERSION
minikube    Ready    control-plane  23d   v1.34.0
1rv24mc012_akanksha@akanshasPC:~/Program10$ kubectl get svc
NAME            TYPE        CLUSTER-IP      EXTERNAL-IP  PORT(S)          AGE
kubernetes      ClusterIP   10.96.0.1       <none>       443/TCP          23d
kubia           NodePort    10.111.41.23    <none>       8080:30236/TCP   10d
nginx-service   NodePort    10.107.62.10    <none>       80:32284/TCP     27s
testdeploy      NodePort    10.102.248.183  <none>       80:32218/TCP     18d
```

### Access Application in Browser

minikube service nginx-service

```
1rv24mc012_akanksha@akanshasPC:~/Program10$ minikube service nginx-service
```

| NAMESPACE | NAME          | TARGET PORT | URL                       |
|-----------|---------------|-------------|---------------------------|
| default   | nginx-service | 80          | http://192.168.49.2:32284 |

🌐 Opening service default/nginx-service in default browser...

🔒 Not Secure http://192.168.49.2:32284

## Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to [nginx.org](http://nginx.org).  
Commercial support is available at [nginx.com](http://nginx.com).

Thank you for using nginx.

Welcome to nginx!

Kubernetes Dashboard

http://127.0.0.1:35841/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard/proxy/#/workloads/namespace=default

Sign in

kubernetes

default

Search

+

🔔

Workloads

Workloads

Cron Jobs

Daemon Sets

Deployments

Jobs

Pods

Replica Sets

Replication Controllers

Stateful Sets

Service

Ingresses

Ingress Classes

Services

Config and Storage

Config Maps

Persistent Volume Claims

Secrets

Storage Classes

Cluster

Cluster Role Bindings

Cluster Roles

Events

Namespaces

Network Policies

Workload Status

Running: 1

Deployments

Running: 3

Pods

Running: 1

Replica Sets

Deployments

| Name             | Images       | Labels     | Pods  | Created ↑      |
|------------------|--------------|------------|-------|----------------|
| nginx-deployment | nginx:latest | app: nginx | 2 / 2 | 14 minutes ago |

Pods

| Name                             | Images       | Labels                                     | Node     | Status  | Restarts | CPU Usage (cores) | Memory Usage (bytes) | Created ↑      |
|----------------------------------|--------------|--------------------------------------------|----------|---------|----------|-------------------|----------------------|----------------|
| nginx-deployment-6f966444b-r5lw7 | nginx:latest | app: nginx<br>pod-template-hash: 6f966444b | minikube | Running | 0        | 0.00m             | 10.54Mi              | 14 minutes ago |
| nginx-deployment-6f966444b-vs9fr | nginx:latest | app: nginx<br>pod-template-hash: 6f966444b | minikube | Running | 0        | 0.00m             | 19.27Mi              | 14 minutes ago |